

Instantaneous Magnetic Field Distribution in Brushless Permanent Magnet dc Motors, Part III: Effect of Stator Slotting

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Abstract—A method for modelling the effect of stator slotting on the magnetic field distribution in the airgap/magnet region of a radial-field brushless permanent magnet dc motor equipped with a surface mounted magnet rotor is presented to enable the analytical prediction of the open-circuit, armature-reaction, and resultant magnetic field distribution under any specified load condition. It is shown that the conventional “1-d” relative permeance model, which is used extensively in the analysis of induction and synchronous machines, is inappropriate for permanent magnet motors, for which the proposed analytical method accounts for stator slot openings by the application of the conformal transformation method and a “2-d” relative permeance function. The results of predictions from the analysis are compared with corresponding finite element analyses.

I. INTRODUCTION

IN Parts I and II [3], [4] of this series of papers analytical models were established for the calculation of the open-circuit and armature-reaction components of magnetic field in the airgap/magnet region of a radial-field permanent magnet brushless dc motor. However they were applicable only to slotless machines having either an internal or an external rotor. By utilizing the concepts of permeance and mmf, applying the conformal transformation method, and introducing a “2-d” relative permeance function, the models can be extended to account for the effects of stator slot openings on the magnet field distribution.

II. CALCULATION METHOD

Although the method to be described is based on a two-dimensional analytical field calculation, to illustrate the philosophy behind the calculation 1-d methods are first reviewed.

A. Simple 1-D Model

From a simple 1-d analysis of the magnetic circuit shown in Fig. 1, in which flux leakage and fringing are neglected, the permeability of the iron yoke is assumed to be infinite, and the magnets are mounted adjacent to the

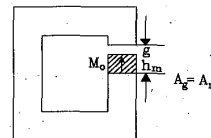


Fig. 1. Simple “1-d” model.

working airgap such that $A_m = A_g$, where A_m and A_g are the cross-sectional areas of the magnet and the airgap, respectively, the airgap flux density B_g is given by

$$B_g = \frac{\mu_o M_0}{1 + \mu_r \frac{g}{h_m}} = \frac{B_r}{g + \frac{h_m}{\mu_r}} = \left(\frac{\mu_o}{g'} \right) (H_c h_m) \quad (1)$$

where $M_0 = B_r / \mu_o$ is the magnetization; B_r is the remanence; H_c is the normal coercivity; μ_r is the relative recoil permeability of the magnet, which is assumed to have a linear demagnetization curve; g is the length of the airgap; and h_m is the thickness of the magnet; and $g' = g + h_m / \mu_r$ is an effective airgap.

If the mmf F is defined as

$$F = H_c h_m = \frac{B_r}{\mu_o \mu_r} h_m \quad (2)$$

and the permeance Λ is defined as:

$$\Lambda = \frac{\mu_o}{g'} \quad (3)$$

then the airgap flux density is given by the product of the permeance and the mmf, i.e.,

$$B_g = F \Lambda \quad (4)$$

Since $(H_c h_m)$ is equivalent to a current source whilst g' is an equivalent effective airgap, the above are consistent with the usual definitions applied to induction and synchronous machines [1].

B. Refined 1-D Model

In Fig. 2(a) the magnets in a brushless dc motor are modelled by equivalent current sheets, which are converted into equivalent current carrying conductors in Fig. 2(b). If the airgap flux density is assumed to vary only

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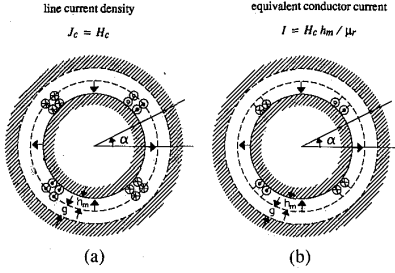


Fig. 2. Refined "1-d" model. (a) Physical model. (b) "1-d" mmf model.

with the circumferential coordinate, α , and to depend on the angle of rotation of the rotor, α_{ma} , as shown in Fig. 2, then from one-dimensional field theory:

$$B_g(\alpha) = \frac{\mu_o M(\alpha, \alpha_{ma})}{1 + \mu_r \frac{g(\alpha)}{h_m}} = \left[\frac{M(\alpha, \alpha_{ma})}{\mu_r} h_m \right] \left[\frac{\mu_o}{g(\alpha) + \frac{h_m}{\mu_r}} \right] = F_\lambda \quad (5)$$

where the mmf F is given by

$$F = \frac{M(\alpha, \alpha_{ma})}{\mu_r} h_m \quad (6)$$

and the permeance Λ is

$$\Lambda = \frac{\mu_o}{g'(\alpha)} \quad (7)$$

where

$$g'(\alpha) = g(\alpha) + \frac{h_m}{\mu_r}$$

If on open circuit the flux paths are assumed to pass radially across the airgap/magnet region and to take circular trajectories into the slot openings, the airgap length $g(\alpha)$ becomes

$$g(\alpha) = g$$

for

$$(k-1)\alpha_t + \frac{\alpha_o}{2} \leq \alpha \leq \alpha_t - \frac{\alpha_o}{2}, \quad k = 1, 2, \dots, Q_s \quad (8a)$$

and

$$g(\alpha) = g + \frac{\pi}{2} R_s \left\{ \frac{\alpha_o}{2} - [|\alpha| - (k-1)\alpha_t] \right\}$$

for

$$(k-1)\alpha_t - \frac{\alpha_o}{2} \leq \alpha \leq (k-1)\alpha_t + \frac{\alpha_o}{2}, \quad k = 1, 2, \dots, Q_s \quad (8b)$$

where $\alpha_o = b_o/R_s$; $\alpha_t = \tau_t/R_s$; b_o is the width of a stator slot opening; τ_t is the stator tooth pitch; R_s is the radius of the stator surface adjacent to the airgap; Q_s is the total number of stator slots; g is the actual airgap length.

C. 2-D Model

In the above 1-d models the magnets have been represented as constant mmf sources, with the pole faces being equi-potential surfaces. In practice, however, because of interpolar flux leakage, the working point of the magnets will not be constant throughout their volume, so that the mmf acting across the airgap will vary over a pole-arc. Hence, as was mentioned in [2], the simplifying assumption of a constant mmf between the two faces of a pole can lead to significant error.

Therefore, in the ensuing analysis, instead of using the mmf and permeance models which were described earlier, the magnetic field is expressed as the flux density B_g produced by the magnets or the stator conductors assuming a smooth airgap as in parts I and II [3], [4], but which is then modulated by the relative permeance $\tilde{\Lambda}$ due to the stator slotting, i.e.,

$$B'_g = B_g \tilde{\Lambda} \quad (9)$$

With this definition (1) for the simple 1-d model can be re-written as

$$B'_g = B_g \tilde{\Lambda}$$

where

$$B_g = \frac{B_r}{1 + \mu_r \frac{g}{h_m}}$$

and

$$\tilde{\Lambda} = 1 \quad (10)$$

while (5) for the refined 1-d model becomes

$$B'_g = B_g \tilde{\Lambda}$$

where

$$B_g = \frac{\mu_o M(\alpha, \alpha_{ma})}{1 + \mu_r \frac{g_o}{h_m}}$$

and

$$\tilde{\Lambda} = \frac{g + \frac{h_m}{\mu_r}}{g(\alpha) + \frac{h_m}{\mu_r}} \quad (11)$$

However, the field deduced from both 1-d models remains the same as before.

In the 2-d model, the calculation of the open-circuit field $B_{open-circuit}$ and the armature-reaction field $B_{armature-reaction}$ proceeds as follows:

- 1) Calculation of $B_{open-circuit}(r, \alpha, \alpha_{ma})$ and $B_{armature-reaction}(r, \alpha, \alpha_{ma})$

The open-circuit field distribution is calculated from the product of the field produced by the magnets when stator

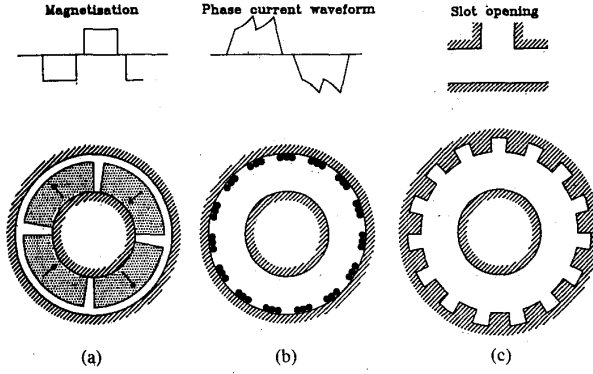


Fig. 3. Features of "2-d" analytical models. (a) Model for calculating magnetic field produced by magnets. (b) Model for calculating magnetic field produced by windings. (c) Model for calculating relative permeance.

slot openings are neglected and the relative permeance function of the slotted airgap region. Similarly, the armature reaction field distribution is calculated from the product of the magnetic field produced by the stator windings when stator slot openings are neglected and the same relative permeance function, i.e.,

$$B_{open-circuit}(r, \alpha, \alpha_{ma}) = B_{magnet}(r, \alpha, \alpha_{ma}) \tilde{\lambda}(r, \alpha) \quad (12a)$$

$$B_{armature-reaction}(r, \alpha, \alpha_{ma}) = B_{winding}(r, \alpha, \alpha_{ma}) \tilde{\lambda}(r, \alpha). \quad (12b)$$

The principle is illustrated in Fig. 3, whilst the calculation of the field components $B_{magnet}(r, \alpha, \alpha_{ma})$ and $B_{winding}(r, \alpha, \alpha_{ma})$ have been described in parts I and II [3], [4]. The calculation of the relative permeance function $\tilde{\lambda}(r, \alpha)$ now follows.

2) Calculation of $\tilde{\lambda}(r, \alpha)$

The permeance $\lambda(r, \alpha)$ is calculated from the flux density distribution $B'(r, \alpha)$ assuming unit difference in magnetic potential between the stator and rotor iron of the slotted machine, shown in Fig. 3(c). Since $\tilde{\lambda}(r, \alpha)$ is defined as the relative permeance with a unity maximum value it is obtained by dividing $\lambda(r, \alpha)$ by a reference permeance λ_{ref} equal to the one-dimensional permeance of an equivalent slotless stator motor, i.e.,

$$\lambda_{ref} = \frac{\mu_o}{g + \frac{h_m}{\mu_r}}. \quad (13)$$

Therefore,

$$\tilde{\lambda}(r, \alpha) = \frac{\lambda(r, \alpha)}{\lambda_{ref}} \quad (14)$$

where $\lambda(r, \alpha)$ equals the flux density distribution $B'(r, \alpha)$ in the magnet/airgap region.

3) Calculation of $B_{magnet}(r, \alpha, \alpha_{ma})$ and $B_{winding}(r, \alpha, \alpha_{ma})$

$B_{magnet}(r, \alpha, \alpha_{ma})$ and $B_{winding}(r, \alpha, \alpha_{ma})$ are defined as the radial components of flux density due to the magnets and the stator windings, respectively, in the equivalent slotless motor, as shown in Figs. 3(a) and (b), for which the steady-state magnet field distribution is obtained by solving the governing 2-d Laplacian-Poissonian equations framed in a polar coordinate system, as described in parts I and II [3], [4]. The permanent magnets are assumed to be radially magnetized and to have a linear demagnetization characteristic. In addition machine end-effects are ignored, and the permeability of the rotor and stator iron is still assumed to be infinite.

III. EFFECT OF SLOTTING

Slotting influences the magnetic field in two ways. Firstly it reduces the total flux per pole, an effect which is usually accounted for by introducing the Carter coefficient K_c into the calculation. Secondly, it affects the distribution of the flux in both the airgap and the magnets.

A. Effect of Slotting on Flux Per Pole

When the stator is slotted, an appropriate Carter coefficient K_c can be included to account for the reduction in flux per pole, with g_e and R_{se} being used instead of g and R_s in the calculation of the field produced by the magnets.

For example, for an internal rotor motor:

$$g_e = g + (K_c - 1) g'$$

and

$$R_{se} = R_s + (K_c - 1) g' \quad (15)$$

where $g' = g + h_m/\mu_r$ is the effective airgap and the Carter coefficient K_c is approximated by

$$K_c = \frac{\tau_t}{\tau_t - \gamma g'} \quad (16)$$

where

$$\gamma = \frac{4}{\pi} \left\{ \frac{b_o}{2g'} \tan^{-1} \left(\frac{b_o}{2g'} \right) - \ln \sqrt{1 + \left(\frac{b_o}{2g'} \right)^2} \right\},$$

and

$$\tau_t = \frac{2\pi R_s}{Q_s}.$$

Similarly, for an external rotor motor:

$$g_e = g + (K_c - 1) g'$$

and

$$R_{se} = R_s - (K_c - 1) g'. \quad (17)$$

However, due to the large effective airgap, which equals the actual airgap plus the magnet thickness, the ratio of the slot opening to the effective airgap is usually small, a

typical value being around 1.01. Therefore the effect of stator slotting on the calculation of the flux per pole is normally negligible, the exception being when a large slot opening is used, so as to reduce the cogging torque for example.

B. Effect of Slotting on the Field Distribution: the Relative Permeance Calculation

The permeance of the slotted airgap/magnet region can be calculated by the conformal transformation method with unit magnetic potential applied between the stator and rotor iron surfaces and no magnets present, and by assuming infinitely deep rectilinear stator slots, as shown in Fig. 4. However such a single slot model is clearly most appropriate for motors having a relatively small number of slots. Nevertheless it may still be acceptable even when the ratio of the airgap length to slot pitch is relatively high. In some circumstances, however, a model consisting of a series of slots may be necessary. In this paper the single-slot model is used, the effect of slotting on the circumferential variation of the airgap field being assumed to be similar to that in an induction machine [1]. However since the effective airgap is much larger in a permanent magnet motor, a "2-d" permeance function is introduced, viz

$$\lambda(\alpha, r) = \begin{cases} \Lambda_o \left[1 - \beta(r) - \beta(r) \cos \frac{\pi}{0.8\alpha_o} \alpha \right] & \text{for } 0 \leq \alpha \leq 0.8 \alpha_o \\ \Lambda_o & \text{for } 0.8 \alpha_o \leq \alpha \leq \alpha_t/2 \end{cases} \quad (18)$$

where $\Lambda_o = \mu_o/g'$ and the function $\beta(r)$ is defined at the axis of a stator slot and depends on the radial position. $\beta(r)$ can be determined by the conformal transformation, and is derived in the Appendix as

$$\beta(r) = \frac{1}{2} \left[1 - \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'} \right)^2 (1 + v^2)}} \right] \quad (19)$$

where v is determined from:

$$y \frac{\pi}{b_o} = \frac{1}{2} \ln \left[\frac{\sqrt{a^2 + v^2} + v}{\sqrt{a^2 + v^2} - v} \right] + \frac{2g'}{b_o} \arctan \frac{2g'}{b_o} \frac{v}{\sqrt{a^2 + v^2}} \quad (20)$$

and

$$a^2 = 1 + \left(\frac{2g'}{b_o} \right)^2$$

with

$$y = \begin{cases} r - (R_s - g - h_m) \\ = r - R_s + g' & \text{for an internal rotor motor} \\ (R_s + g + h_m) - r \\ = R_s + g' - r & \text{for an external rotor motor} \end{cases} \quad (21)$$

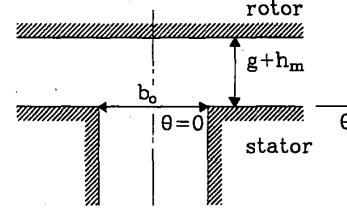


Fig. 4. Permeance calculation model.

The relative permeance function is then calculated from

$$\tilde{\lambda}(\alpha, r) = \frac{\lambda(\alpha, r)}{\Lambda_o} = \frac{\lambda(\alpha, r)}{\mu_o/g'} \quad (22)$$

which can be expressed in the form of a Fourier series, viz

$$\tilde{\lambda}(\alpha, r) = \sum_{\mu=0}^{\infty} \tilde{\Lambda}_{\mu}(r) \cos \mu Q_s (\alpha + \alpha_{sa}) \quad (23)$$

where α_{sa} is determined by the winding pitch, as will be described in part IV [5], and $\tilde{\Lambda}_{\mu}(r)$ can be deduced as:

$$\tilde{\Lambda}_o(r) = \frac{1}{K_c} \left(1 - 1.6\beta \frac{b_o}{\tau_t} \right) \quad (24)$$

and

$$\begin{aligned} \tilde{\Lambda}_o(r) &= \frac{2}{\alpha_t} \int_{-\alpha_t/2}^{\alpha_t/2} \tilde{\lambda}(r, \alpha) \cos \mu Q_s \alpha d\alpha \\ &= \frac{4}{\alpha_t} \left\{ \int_0^{0.8\alpha_o} \left(1 - \beta(r) - \beta(r) \cos \frac{\pi}{0.8\alpha_o} \alpha \right) \right. \\ &\quad \cdot \cos \frac{2\pi}{\alpha_t} \mu \alpha d\alpha + \int_{0.8\alpha_o}^{0.5\alpha_t} \cos \frac{2\pi}{\alpha_d} \mu \alpha d\alpha \left. \right\} \\ &= -\beta(r) \frac{4}{\pi\mu} \left[0.5 + \frac{\left(\mu \frac{b_o}{\tau_t} \right)^2}{0.78125 - 2 \left(\mu \frac{b_o}{\tau_t} \right)^2} \right] \\ &\quad \cdot \sin \left(1.6 \pi \mu \frac{b_o}{\tau_t} \right). \end{aligned} \quad (25)$$

In (24), the Carter coefficient K_c has been included to account for the increase in effective airgap due to the stator slotting.

It is also worth mentioning that since induction motors typically have an airgap of only 0.25–0.3 mm the magnitude of the flux density varies only slightly in passing across the airgap from the stator to the rotor. In other words, the effect of slotting on the distribution of the airgap field is almost independent of radius. Therefore the radial variation of flux density is often ignored [1], i.e., by setting $v = 0$ in (19). However, in the case of permanent magnet motors having magnets mounted adjacent to the airgap, although the actual airgap g is still relatively small, the equivalent airgap $g' = g + h_m/\mu_r$ can be rel-

atively large since the relative recoil permeability μ_r of the magnets is approximately 1 for most magnet materials. As a consequence, in passing radially from the midpoint of a stator slot opening to the rotor the airgap flux density typically varies from a value of around half the maximum at the stator bore to a maximum at the surface of the rotor hub.

IV. COMPARISON WITH THE FINITE ELEMENT CALCULATIONS

The refined 2-d analytical models developed for the open-circuit and armature-reaction field calculation are now validated by comparing predictions with results from corresponding finite element calculations for the internal rotor 3-phase 4-pole brushless dc motor whose parameters are given in part I [3].

A. Relative Permeance Model

Based on the proposed "2-d" relative permeance model, Fig. 5 shows the variations of the relative permeance waveform with radius, whilst Fig. 6 compares the variation of the minimum relative permeance (at the slot centres) with radius with results from both the "1-d" model, which can be obtained from (19) by setting β to a fixed value given by (38) in the Appendix, and the refined "1-d" model. Clearly, stator slotting has a significant effect on the permeance waveform at the stator surface, whilst at the rotor surface the effect is negligible. The pronounced variation of the relative permeance with radius confirms the unsuitability of the 1-d permeance models for permanent magnet motors.

B. Magnet Field Distribution

Fig. 7 shows the finite element discretisation of the internal rotor motor, whilst Fig. 8 compares predictions of the open-circuit flux density distribution at different radii in the magnet/airgap regions by both the proposed analytical method and finite element analysis. Fig. 9 makes similar comparisons for the armature-reaction field distribution. Both show good agreement, and the discussions of part I and part II [3], [4] are equally applicable to the slotted motor.

However, while the results from the analytical technique and the finite element method are in good agreement overall, both in amplitude and waveform, local differences are apparent. For example, in the variation of flux density with radius at an angular position corresponding to the edge of a stator tooth, when the finite element calculation predicts an increase in flux density at the tooth tips, because of their flux concentrating effect, which is absent in the results from the analytical model, primarily due to the fact that a simple permeance waveform has been assumed so as to simplify the calculation. However the analytical calculation could be improved by using a more refined permeance distribution derived from the conformal transformation method, i.e., by relaxing the waveform assumption described by (18) and calculating the

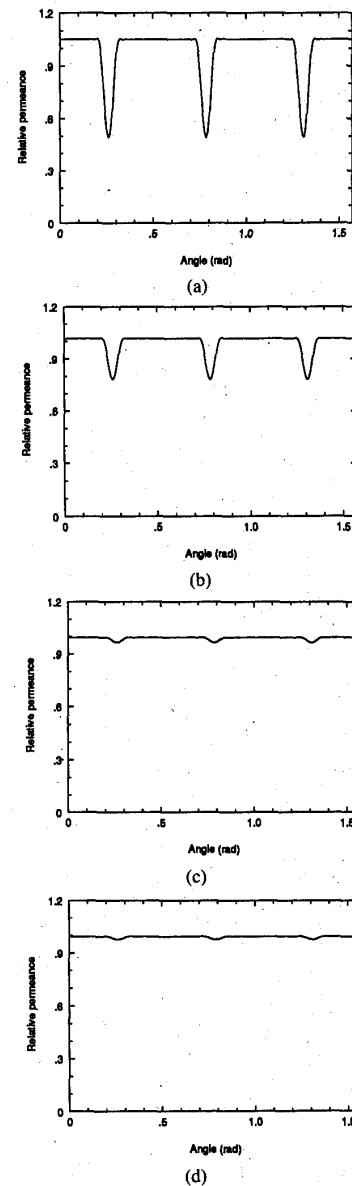


Fig. 5. Variation of relative permeance with radius. (a) $r = R_s$. (b) $r = R_s - g$. (c) $r = R_s - g - 0.5h_m$. (d) $r = R_s - g - h_m$.

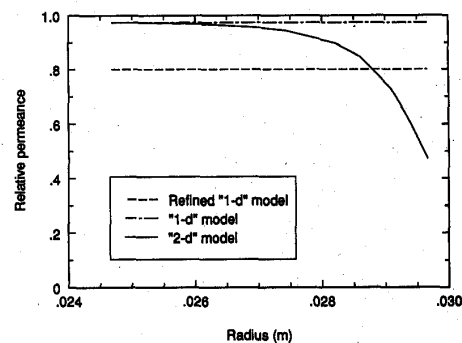


Fig. 6. Variation of the minimum permeance with radius.

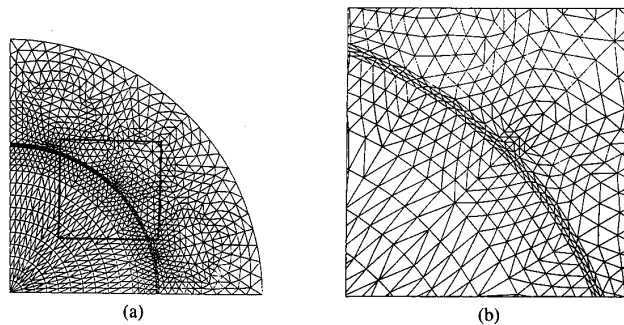
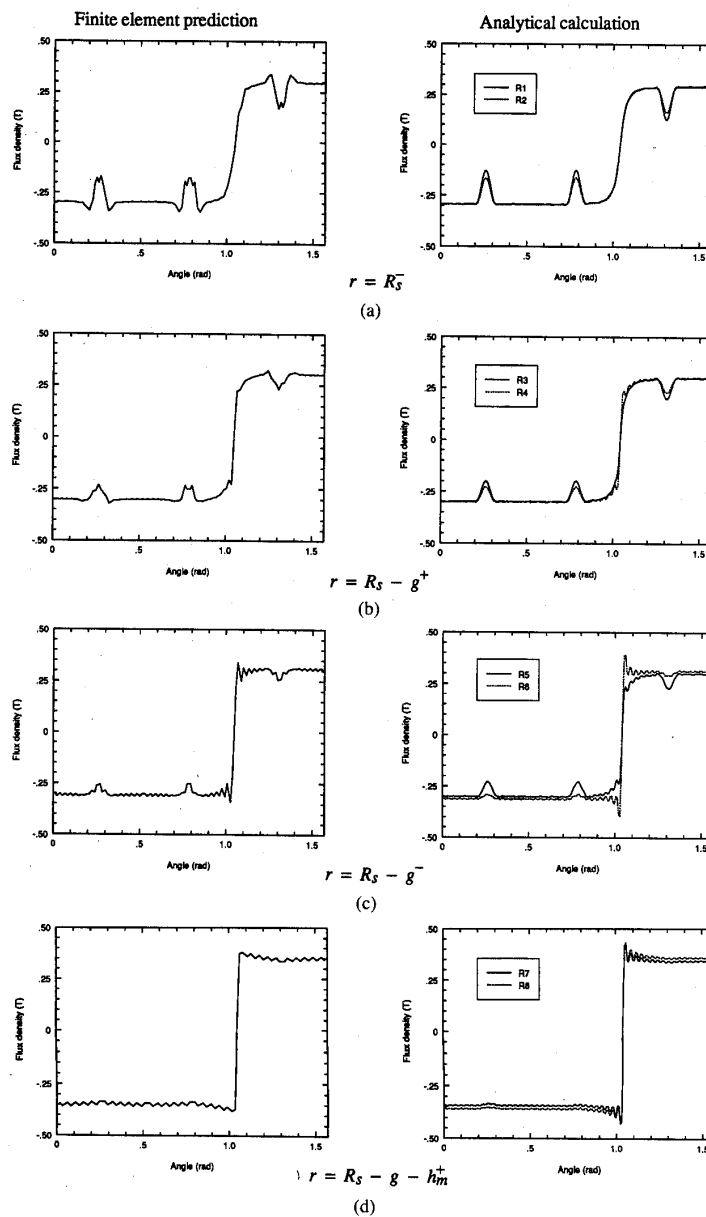


Fig. 7. Finite element mesh. (a) Mesh. (b) Zoom of mesh.

Fig. 8. Comparison of open-circuit field distribution at different radii. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of the radii is given in Table I of Part I [3].

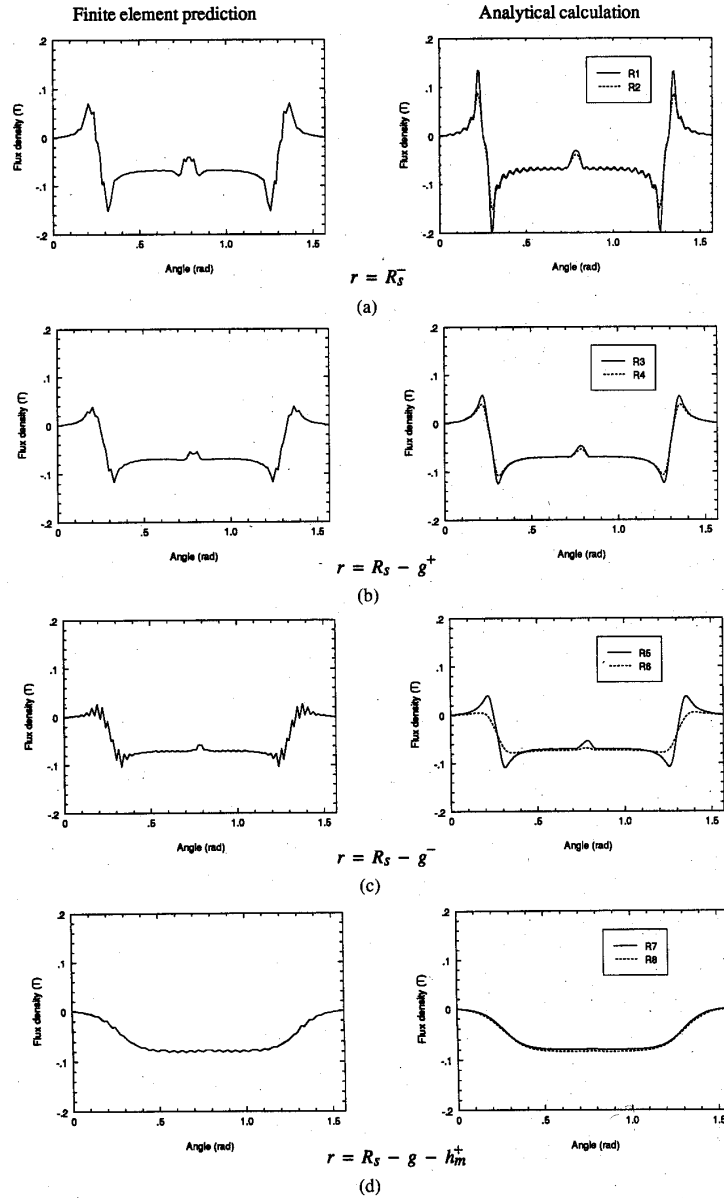


Fig. 9. Comparison of armature reaction field distribution at different radii. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of the radii is given in Table I of Part I.

waveform directly from the field distribution described by (30) in the Appendix. However this would increase the computing time significantly, and, in most cases, is not likely to be critical.

V. CONCLUSIONS

A two-dimensional analytical technique for calculating the magnetic field distribution produced by the permanent magnets and the stator windings of a brushless dc motor has been described. The effect of stator slotting is accounted for incorporating a "two-dimensional" relative permeance function into the analyses which were de-

scribed in parts I and II [3], [4]. The analytically predicted results have been shown to be in good agreement with finite element analyses.

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APPENDIX DETERMINATION OF $\beta(r)$

Based on the idealized single-slot model of Fig. 10, the interior airgap region in the complex plane z is first transformed to the upper half of the complex plane w , and sub-

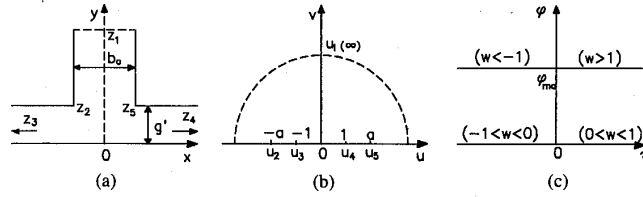


Fig. 10. Conformal transformation. (a) z -plane. ($g' = g + h_m/\mu_r$). (b) w -plane. (c) t -plane.

sequently to the complex plane t , by conformal transformation.

The first transformation is given by

$$z = \frac{b_o}{\pi} \left\{ \arcsin \frac{w}{a} + \frac{g'}{b_o} \ln \left[\frac{\sqrt{a^2 - w^2} + \frac{2g'}{b_o} w}{\sqrt{a^2 - w^2} - \frac{2g'}{b_o} w} \right] \right\} \quad (26)$$

which after some deduction becomes

$$z = \frac{b_o}{\pi} \left\{ \frac{j}{2} \ln \left[\frac{1 - j \frac{w}{\sqrt{a^2 - w^2}}}{1 + j \frac{w}{\sqrt{a^2 - w^2}}} \right] + j \frac{2g'}{b_o} \cdot \arctan \left[\frac{-j 2g' w}{b_o \sqrt{a^2 - w^2}} \right] \right\} \quad (27)$$

where j is the complex operator, and

$$a = \sqrt{1 + \left(\frac{2g'}{b_o} \right)^2}. \quad (28)$$

The second transformation is

$$t = \frac{\varphi_{mo}}{\pi} \ln \frac{1 + w}{1 - w} \quad (29)$$

where $\varphi_{mo} = (B_r/\mu_o\mu_r) h_m$ is the magnetic potential between the stator and rotor iron surfaces. The flux density in the z -plane is then given by

$$\begin{aligned} |B_z| &= \mu_o |H_z| = \mu_o \left| \frac{dt}{dz} \right| = \mu_o \left| \frac{dt}{dw} \right| \left| \frac{dw}{dz} \right| \\ &= \mu_o \frac{\varphi_{mo}}{g'} \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'} \right)^2 - \left(\frac{b_o}{2g'} w \right)^2}}. \end{aligned} \quad (30)$$

By letting $w = \pm 1$, the maximum flux density $B_{z \max}$ at $x = \pm \infty$ is obtained as:

$$B_{z \max} = \mu_o \frac{\varphi_{mo}}{g'}. \quad (31)$$

The variation of flux density against circumferential angular position exhibits a minimum at the axis of the stator slots, the value depending on the radius, and is obtained by letting $w = jv$, i.e.,

$$\begin{aligned} B_{z \min} &= \mu_o \frac{\varphi_{mo}}{g'} \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'} \right)^2 + \left(\frac{b_o}{2g'} v \right)^2}} \\ &= B_{z \max} \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'} \right)^2 (1 + v^2)}} \end{aligned} \quad (32)$$

and

$$\begin{aligned} \beta(r) &= \frac{B_{z \max} - B_{z \min}}{2B_{z \max}} \\ &= \frac{1}{2} \left[1 - \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'} \right)^2 (1 + v^2)}} \right] \end{aligned} \quad (33)$$

v is determined by solving (27) with $z = jy$ and $w = jv$

$$\begin{aligned} jy &= \frac{b_o}{\pi} \left\{ \frac{j}{2} \ln \left[\frac{1 - j \frac{v}{\sqrt{a^2 + v^2}}}{1 + j \frac{v}{\sqrt{a^2 + v^2}}} \right] + j \frac{2g'}{b_o} \cdot \arctan \left[\frac{-j 2g' (jv)}{b_o \sqrt{a^2 + v^2}} \right] \right\} \end{aligned} \quad (34)$$

therefore

$$\begin{aligned} y \frac{\pi}{b_o} &= \frac{1}{2} \ln \left[\frac{\sqrt{a^2 + v^2} + v}{\sqrt{a^2 + v^2} - v} \right] \\ &+ \frac{2g'}{b_o} \arctan \frac{2g' v}{b_o \sqrt{a^2 + v^2}} \end{aligned} \quad (35)$$

where y is given by

$$y = \begin{cases} r - (R_s - g - h_m) \\ \quad = r - R_s + g' & \text{for internal rotor motor} \\ (R_s + g + h_m) - r \\ \quad = R_s + g' - r & \text{for external rotor motor} \end{cases} \quad (36)$$

At the rotor surface $B_{z \min}$ is determined by setting $v = 0$, to give

$$B_{z \min} = B_{z \max} \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'}\right)^2}} \quad (37)$$

Hence

$$\beta(r)|_{r=R_r} = \frac{1}{2} \left[1 - \frac{1}{\sqrt{1 + \left(\frac{b_o}{2g'}\right)^2}} \right] \quad (38)$$

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